Do microbial communities separate by habitat? A re-analysis of the Global Patterns 16S dataset

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1 Summary

The Global Patterns dataset (Caporaso et al., 2011) sequenced the 16S rRNA gene from environmental and host-associated samples, and the original paper reported that communities split between host-associated and free-living habitats. Using 23 non-control samples (18,773 OTUs; three mock-community controls excluded), I test that split directly. Habitat (host-associated vs free-living) explains 12% of Bray-Curtis dissimilarity and 22% of weighted-UniFrac distance (PERMANOVA; $p = 0.001$ for both, the smallest value 999 permutations can give), while the finer sample type explains 69% and 77%. Bray-Curtis on unrarefied relative abundances gives the same answer (12%, $p = 0.001$). At phylum level, DESeq2 (Love et al., 2014) finds Firmicutes, Tenericutes and Fusobacteria strongly enriched in host-associated samples. Because both of those choices are contested, I re-ran the analysis without them: the habitat split holds on an Aitchison distance with no rarefying at all (14%, $p = 0.001$), and the three phyla survive ANCOM-BC2 (Lin & Peddada, 2020), but ANCOM-BC2 calls only 5 of 34 phyla where DESeq2 calls 21, so that count is a property of the method rather than of the data. With 2 to 4 samples per sample type, the tests are underpowered for any single type, and the dispersion assumption behind PERMANOVA is borderline (see "Is the PERMANOVA result trustworthy?").

2 Question and data

Do the microbial communities of different habitats group by habitat, and which taxa distinguish host-associated from free-living communities? The data are the `GlobalPatterns` object distributed with the Bioconductor package `phyloseq` (McMurdie & Holmes, 2013): an OTU table with 19,216 OTUs and 26 samples, a taxonomy table and a phylogenetic tree. Sample types and counts after removing the mock communities: Feces (4), Freshwater (2), Freshwater (creek) (3), Ocean (3), Sediment (estuary) (3), Skin (3), Soil (3), Tongue (2). Sequencing depth ranges from 58,688 to 2,357,181 reads per sample.

3 Methods

Rarefaction. For alpha and beta diversity, samples were subsampled without replacement to the smallest library (58,688 reads), which leaves 12,386 OTUs. Rarefying throws away data and is criticised for differential abundance testing (McMurdie & Holmes, 2014), so I use it only for diversity, run a non-rarefied sensitivity check, and never for the DESeq2 analysis.

Alpha diversity. Observed OTUs and Shannon index. Sample types compared with a Kruskal-Wallis test; habitats with a Wilcoxon rank-sum test.

Beta diversity. Bray-Curtis dissimilarity and weighted UniFrac distance, ordinated by principal coordinates analysis (PCoA). Group differences tested by PERMANOVA (`vegan::adonis2`, 999 permutations) with habitat or sample type as the single predictor. Homogeneity of dispersion was checked with `betadisper`, because PERMANOVA can return a significant result from unequal spread alone.

Differential abundance. Counts were aggregated to phylum, phyla with fewer than 1,000 total reads dropped, and DESeq2 fitted with `~ habitat` (free-living as reference) using `poscounts` size factors, which suit sparse count

tables. Adjusted $p < 0.05$ (Benjamini-Hochberg).

Software. R 4.5.3, phyloseq 1.54.2, vegan 2.7.3, DESeq2 1.50.2.

4 Results

4.1 Sequencing depth and alpha diversity

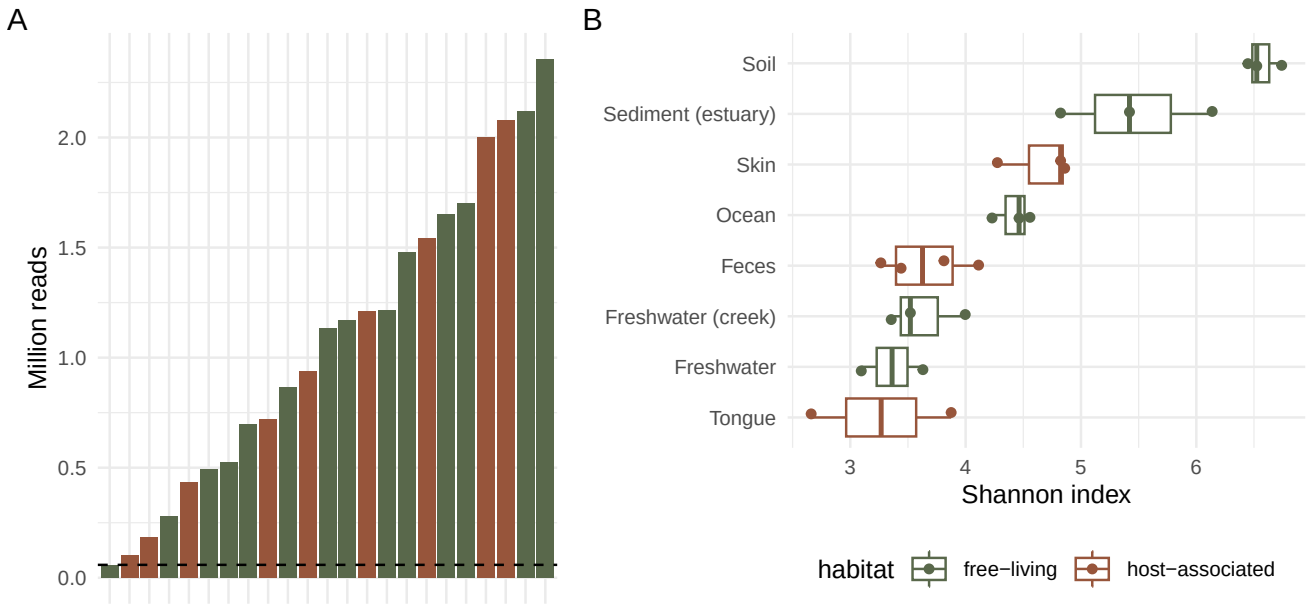


Figure 1: (A) Reads per sample before rarefaction (dashed line = rarefaction depth, barely visible at this scale). (B) Shannon diversity after rarefaction, by sample type.

Library sizes vary about 40-fold, from 58,688 to 2,357,181 reads (Figure 1A), which is why diversity is compared at a common depth. Mean Shannon diversity is 4.78 for free-living and 3.9 for host-associated samples (Wilcoxon $p = 0.16$, not significant at this sample size). Shannon diversity differs among the 8 sample types in a Kruskal-Wallis test ($p = 0.0076$), but with 2 to 4 samples per type, pairwise comparisons would be unreliable and I do not report them.

4.2 Community composition

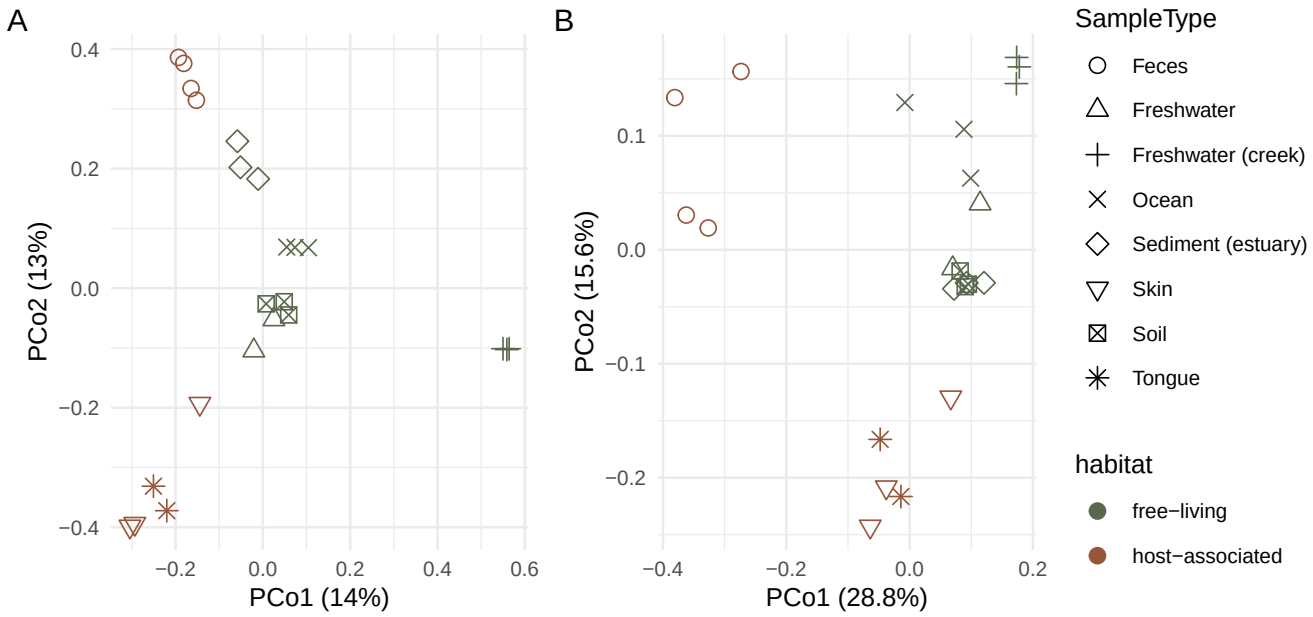


Figure 2: PCoA of (A) Bray-Curtis dissimilarity and (B) weighted UniFrac distance on rarefied data. Colour = habitat, shape = sample type.

Table 1: PERMANOVA (999 permutations). R^2 is the share of total dissimilarity explained by the predictor.

Distance	Predictor	R^2	Pseudo-F	p
Bray-Curtis	Habitat (2 groups)	0.119	2.84	0.001
Bray-Curtis	Sample type (8 groups)	0.690	4.78	0.001
Weighted UniFrac	Habitat (2 groups)	0.216	5.80	0.001
Weighted UniFrac	Sample type (8 groups)	0.769	7.14	0.001

Samples of the same type mostly sit together in both ordinations (Figure 2; one skin sample sits apart from the other two), and sample type explains most of the variation (Table 1). Habitat alone explains less, because the free-living group contains very different environments (soil, ocean, freshwater, sediment). Weighted UniFrac, which accounts for phylogenetic relatedness and abundance, gives a larger habitat effect than Bray-Curtis (22% against 12%). That is consistent with differences in deeper lineages as well as in which OTUs are present, but I did not test that directly.

4.3 Is the PERMANOVA result trustworthy?

PERMANOVA assumes similar within-group spread. Dispersion differs between habitats at $p = 0.065$ (`betadisper`, Bray-Curtis), which is borderline: part of the Bray-Curtis habitat difference could reflect unequal spread rather than a shift in centroid. Two things limit the worry. The weighted-UniFrac result, which is less sensitive to rare OTUs, is stronger, and the non-rarefied relative-abundance version gives $R^2 = 0.12$ ($p = 0.001$). Neither removes the caveat; it is a reason to describe the effect as a difference in composition and dispersion together.

4.4 Which phyla distinguish host-associated communities?

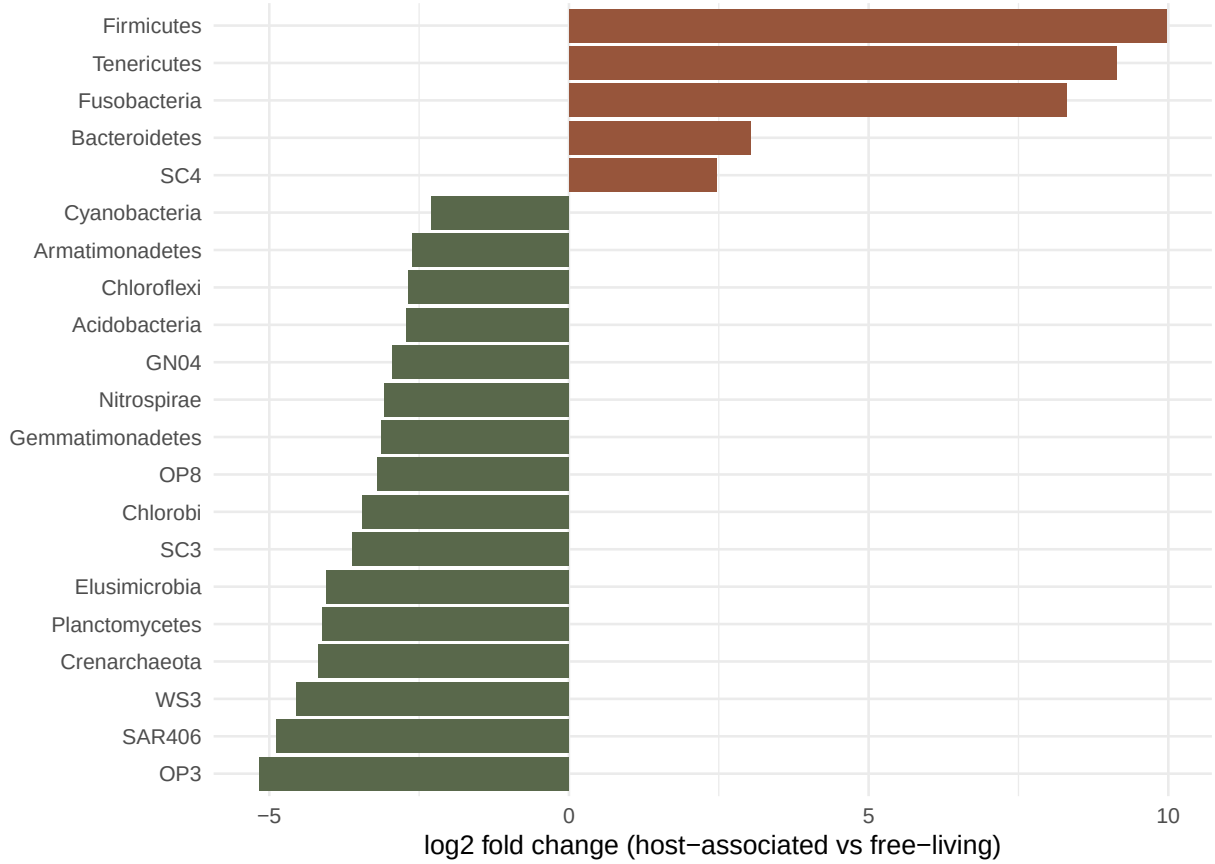


Figure 3: Phyla with adjusted p < 0.05: log2 fold change in host-associated relative to free-living samples (DESeq2, poscounts size factors).

Of 34 phyla tested, 21 differ at adjusted $p < 0.05$: 5 are higher in host-associated samples and 16 higher in free-living samples (Figure 3). The strongest host-associated enrichments are Firmicutes, Tenericutes and Fusobacteria. The host-associated group is only gut, skin and tongue (9 samples). Log2 fold changes as large as the ones seen here (up to about 10) usually arise when a phylum is nearly absent in one group; I did not check that for each phylum, so I read the ranking as qualitative and the fold changes as imprecise.

4.5 Does the result survive without rarefying, and without DESeq2?

Both of the choices above are contested. McMurdie & Holmes (2014) argue that rarefying discards usable data, and the compositional school (Gloor et al., 2017) argues that a count table carries only relative information, so a method that does not model each sample's sampling fraction will call too many taxa. Rather than defend the choices, I re-ran both questions the other way (`R/compositional.R`).

Beta diversity without rarefying. A centred log-ratio (CLR) transform with Aitchison distance uses every read. On OTUs seen in at least 3 samples (12,267 of 18,773 OTUs, 99.7% of reads), habitat explains 14% of Aitchison distance ($p = 0.001$) and sample type 78%. That sits between the rarefied Bray-Curtis estimate (12%) and weighted UniFrac (22%), and it is stable across the prevalence filter (13%, 14%, 14% for OTUs in at least 2, 3, 5 samples). **The habitat split does not depend on rarefying.**

It does, however, sharpen the dispersion caveat rather than settling it: `betadisper` on the Aitchison distance gives $p = 0.0022$, against 0.065 on Bray-Curtis. Spread differs between habitats more clearly on this distance, not less.

Table 2: Taxa called differentially abundant between habitats at adjusted $p < 0.05$ by DESeq2 with poscounts size factors and by ANCOM-BC2, on the same unrarefied counts.

Rank	Taxa tested	DESeq2	ANCOM-BC2	Both	DESeq2 only	ANCOM-BC2 only	Effect-size r
Phylum	34	21	5	5	16	0	0.911
Genus	292	208	75	70	138	5	0.944

Differential abundance against a bias-corrected method. ANCOM-BC2 (Lin & Peddada, 2020) models the sampling fraction explicitly. On the same counts it calls far fewer taxa than DESeq2: 5 of 34 phyla against DESeq2's 21, and 75 of 292 genera against 208 (Table 2). The two methods nonetheless agree closely on *effect size* (Pearson $r = 0.911$ at phylum, 0.944 at genus); they disagree on which effects clear a significance threshold. At phylum level every ANCOM-BC2 call is inside DESeq2's set; at genus level 5 genera are called by ANCOM-BC2 alone.

The three phyla I named above survive both methods (Firmicutes, Tenericutes, Fusobacteria), so the headline reading of this dataset stands. The *count* does not: "21 of 34 phyla differ" is a statement about DESeq2, not about the data. ANCOM-BC2 warns that its bias estimation wants more than 50 taxa, which the phylum table does not have, but the genus analysis has 292 and shows the same pattern, so the gap is not an artefact of too few taxa.

5 Discussion and limitations

Replication is very small. There are 2 to 4 samples per sample type and 9 versus 14 per habitat. Group-level tests are valid in principle but have little power, so non-significant results here are not evidence of no difference.

The groups are not independent replicates. The habitats combine different environments, and samples from the same site type may share a source. The test treats each sample as independent evidence for "host-associated vs free-living", which overstates how much independent information the data contain.

Rarefaction. Used only for diversity, with a non-rarefied check. Dropping to 58,688 reads per sample discards most of the data for deeply sequenced samples. The Aitchison analysis avoids it entirely; it needs a pseudocount for the zeros instead, which is its own arbitrary choice, and I did not compare pseudocount strategies.

Phylum level. Aggregating to phylum makes the count table well-behaved for DESeq2 but hides the OTU- and genus-level changes that usually matter in practice.

Scope. This reproduces a qualitative finding of the original paper on a dataset already prepared in R; it does not involve the raw reads, OTU picking or taxonomy assignment. What it adds to the original is the method comparison: the habitat result is shown to survive dropping rarefaction and switching to a bias-corrected differential-abundance test, while the number of taxa called is shown not to.

6 Provenance and reproducibility

The data are public and shipped with Bioconductor `phyloseq` (from Caporaso et al. (2011)). The analysis code is this document (`report/report.qmd`). It was written and run with the assistance of Claude Code (Anthropic); I am responsible for checking and understanding every result before relying on it. No part derives from university coursework or assessed material.

References

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